

Arjun Krishnan

Software engineer specializing in graphics and game engine programming

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Education

University of Southern California

August 2026 - May 2028

Master of Science in Computer Science - Game Development

University of California, Santa Cruz

September 2022 - December 2025

Bachelor of Science in Computer Science - Game Design, Minor in Computer Science. GPA: 3.97

Projects

Hyacinth Game Engine & Server - C++, Vulkan, GLSL, PhysX

January 2026 - Current

Deferred-rendered, multiplayer first-person shooter game engine and server.

- Engineered a multithreaded authoritative game server to connect game clients. Features a 128-hertz tickrate, client prediction/server reconciliation, and lag compensation.
- Developed a volume-based Dynamic Diffuse Global Illumination (DDGI) system using ray tracing to calculate indirect illumination. Modeled after Overwatch's baked DDGI system; optimized with a stencil buffer to ensure one evaluation per pixel.
- Implemented hardware-traced specular reflections, stencil-buffered outlines, FXAA, third-person locomotion, and a UI system that supports elements in world space.

Stingray Raytracer - C++, CUDA

June 2023 - September 2023

Real-time ray-tracing application, accelerated on the GPU using NVIDIA's CUDA API.

- Incorporated iterative anti-aliasing, physically based materials, and area light sampling for soft shadows.
- Optimized frame time from **350 ms** to **15 ms (~23x speedup)** by implementing bounding volume hierarchies and parallelizing ray calculations on the GPU.
- Capable of ray tracing models with **over 850,000 primitives** in real time.

Orchid Game Engine - C++, Vulkan, GLSL, PhysX

June 2024 - July 2025

Forward-rendered 3D game engine, blending elements of photorealistic and non-photorealistic rendering.

- Programmed graphics techniques including compute skinning, PBR texture processing, cascaded shadow mapping, physically based bloom, inverse hull outlines, and a custom toon shader.
- Created singleton managers for graphics, physics, time, and input, facilitating object scripting and animation. Enabled rapid prototyping of games with unique mechanics.
- Utilized frame analysis tools such as RenderDoc and NVIDIA Nsight to locate bottlenecks. Reduced frame time from **8.1 ms/frame** to **6.15 ms/frame (32%)** through compute frustum culling and a depth prepass.

Work Experience

Sony Interactive Entertainment: PlayStation - Software Engineering Intern

Summer 2025

- Engineered a dynamic UI navigation system for menus in a cross-platform, multiplayer game built in Unreal Engine 5, speeding up iteration and testing for developers.
- Designed, prototyped, and polished a proof-of-concept Match-3 mobile game using Godot.
- Built a particle state machine, a state persistence manager, and a tutorial card framework; incorporated official Astro Bot assets sourced from designers on the mobile development team.
- Integrated the game into the PlayStation App as a landing screen tile using React Native and JavaScript.
- Wrote a custom domain warping shader and a .OBJ file loader using the PS5 Graphics SDK to render on a console devkit. Optimized the loader by tightly packing structs to reduce memory footprint.

Technical Skills

Languages/Tools: C++, Vulkan, CUDA, RenderDoc, NVIDIA Nsight, GLSL, GLM, Dear ImGui, Git, Jenkins